

MOHAMMAD NAVED

Lead AI Engineer and Full-Stack Product Architect | Founder, Codtrex AI

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PROFILE SUMMARY

Founder and Lead AI Engineer with a proven track record of shipping production-grade, full-stack products across IoT, SaaS, and AI domains. Founded Codtrex AI, a boutique engineering studio delivering custom LLM integrations, cloud systems, and smart hardware solutions for clients. Built and achieved official Google Smart Home certification for 4Layers, a custom ESP32-based IoT ecosystem connected to AWS IoT Core and Google Home. Also shipped MediStock, a live B2B cloud pharmacy SaaS used by pharmacies. Deep expertise spanning embedded firmware (ESP32, Arduino/C++), cloud infrastructure (AWS, Docker), modern web development (React 19, Next.js 15/16, TypeScript), and AI engineering (Gemini API, multi-LLM orchestration). Seeking senior engineering roles and consulting engagements where full-stack ownership and product velocity are core to the mission.

CORE EXPERTISE

- **AI and LLM Engineering** - production Gemini API integration, multi-LLM orchestration, agentic workflows, prompt architecture, RAG system design
- **Full-Stack Product Development** - React 19, Next.js 15/16, TypeScript, Tailwind CSS v4, Vite, Supabase (PostgreSQL), FastAPI (Python)
- **IoT and Embedded Systems** - ESP32 firmware (Arduino/C++), MQTT protocol, AWS IoT Core, mTLS device authentication, OTA firmware updates
- **Cloud and DevOps** - AWS (IoT Core, App Runner, ECR, S3), Docker, Vercel, GitHub Actions CI/CD pipelines
- **Mobile Development** - React Native (Expo), Capacitor Android for Google Play publishing
- **Studio and Client Delivery** - end-to-end product ownership from requirements to deployment; Google Smart Home certification lifecycle

EXPERIENCE

Codtrex AI - Boutique AI Engineering Studio2025 - Present

Lead Engineer and Founder | [codtrex.vercel.app](#) | React 19, Vite, Tailwind CSS v4, Supabase, Gemini API, Vercel

- Founded and operate a boutique AI engineering studio offering end-to-end LLM integration, full-stack web products, IoT systems, and technical consultation to clients
- Built and shipped the studio public website with an AI chat assistant, service tiers, and client case studies deployed at [codtrex.vercel.app](#)
- Designed multi-LLM product architectures and delivered client-facing AI solutions across web, mobile, and embedded platforms

4Layers - Google Certified Smart Home IoT Ecosystem2025 - Present

Lead Systems Engineer and Founder | ESP32, AWS IoT Core, MQTT, mTLS, React Native, FastAPI, Docker, AWS App Runner

- Achieved official Google Smart Home certification - passed 71 out of 75 tests in the Google Home Test Suite; devices listed and functional in the Google Home ecosystem
- Architected 3-source real-time state synchronization (ESP32 hardware to AWS IoT Core MQTT to Google Home) achieving zero state divergence across physical buttons, app controls, and voice commands
- Engineered QA Hold Mode to prevent Google Home from reverting local hardware states during certification test runs - a critical blocking issue independently discovered and resolved
- Built OTA 2.0 with version pinning and rollback safety for wireless firmware updates across all deployed devices without physical access
- Designed custom ESP32 hardware: 4-channel relay switch boards and Triac-based multi-speed fan regulator; mTLS certificates provisioned per device via AWS IoT Core

MediStock - Cloud Pharmacy Management SaaS2025 - Present

Lead Engineer and Founder | React 18, Supabase, PostgreSQL, Capacitor Android, Vercel | [medistock-pharma.vercel.app](#)

- Built and shipped a full B2B pharmacy SaaS covering inventory management, expiry tracking, billing, and supplier management for pharmacy businesses
- Deployed cross-platform as a live web application and Android APK via Capacitor: targeting Google Play Store submission

PROJECTS

UsedTech Market - C2C Electronics Marketplace2025

Product Architect | Next.js 15/16, React 19, TypeScript, Supabase, Tailwind CSS v4

- Architected a consumer-to-consumer marketplace for buying and selling used electronics with real-time listings, advanced search filters, and seller dashboards
- Built on Next.js 15/16 App Router with server-side rendering for SEO-optimized product pages and Supabase real-time subscriptions for live listing updates
- Designed the full product data model, authentication flow, and listing lifecycle from draft to published to sold states

AutoApply AI - Automated Job Application System2025

Self-Initiated | Python, Gemini AI, Multi-LLM Orchestration, Prompt Engineering

- Designed a multi-step prompt chain that reads job descriptions, extracts key requirements, and generates tailored resumes and cover letters through structured LLM calls
- Engineered retry and fallback prompt logic to handle malformed or incomplete model outputs ensuring consistent output quality in production use
- Built output validation layer to enforce format, tone, and length constraints across hundreds of varying job postings

DataLens AI - Full-Stack AI Analytics Platform2025

Self-Initiated | React, TypeScript, Recharts, Gemini AI, DuckDB, SQL

- Built an AI analytics platform with a natural-language-to-SQL chat interface enabling non-technical users to query data through plain English questions
- Developed ML Lab module for model output explanation and business insight surfacing using Gemini-powered prompt templates
- Designed the ETL pipeline, SQL engine, and data visualization layer end-to-end as a single engineer

Interactive Portfolio and PDF Automation Pipeline2025

Self-Initiated | React 19, TypeScript, Framer Motion, Tailwind CSS v4, Node.js, Puppeteer, Gemini AI

- Built a single-page interactive portfolio with hardware-accelerated animations, an AI chatbot powered by Gemini, and a context-aware navigation system
- Automated resume PDF generation via a Node.js and Puppeteer script producing a pixel-perfect A4 PDF from live HTML source, eliminating manual export steps from the update workflow

TECHNICAL SKILLS

AI and LLM: Gemini API, Multi-LLM Orchestration, Prompt Engineering, Agentic Workflows, RAG Systems

Frontend: React 19, Next.js 15/16, TypeScript, Tailwind CSS v4, Vite, Framer Motion, Recharts

Backend: FastAPI (Python), Node.js, REST APIs, Supabase, PostgreSQL, SQL, DuckDB

IoT and Embedded: ESP32 (Arduino/C++), MQTT, AWS IoT Core, mTLS, OTA Firmware Updates, Triac control

Cloud and DevOps: AWS IoT Core, AWS App Runner, AWS ECR, AWS S3, Docker, Vercel, GitHub Actions

Mobile: React Native (Expo), Capacitor Android

Languages: TypeScript, Python, C++ (Arduino), SQL, JavaScript

EDUCATION

B.Tech, Computer Science - RGPV University, Indore2025

Rajiv Gandhi Proudhyogiki Vishwavidyalaya | Madhya Pradesh, India

CERTIFICATIONS AND ACHIEVEMENTS

- Google Smart Home Certification - 4Layers IoT Ecosystem (71 out of 75 tests passed in the Google Home Test Suite)
- Generative AI with Gemini - Prompt Engineering and LLM Application Development
- Python for Data Analysis and Visualization - Self-directed
- SQL for Data Analytics - Relational Database Workflows
- Power BI - Interactive Dashboard Design and Reporting